TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Determination of the Moment of Inertia of a Flywheel

A falling mass trades its potential energy for the wheel's rotation — timing the fall and counting the coast-down turns separates the wheel's inertia from friction.

PHY-1104 · PHYSICS-I SESSIONAL · EXPERIMENT 2 · LEVEL 1 TERM 1

1. Aim and theory

Aim: to find the moment of inertia I of a flywheel about its axle. A mass m hangs from a cord wrapped round the axle (radius r). Released from rest, it falls a measured height h in time t, spinning the wheel up; the cord then leaves the axle and the wheel coasts through N further revolutions before friction stops it. Energy bookkeeping over the fall, with the coast-down used to estimate the friction work per revolution, gives the working formula in its standard laboratory form:

$$I = m\,r^2\left(\frac{g\,t^2}{2h} - 1\right) \times \left(\frac{N}{N+n}\right) \quad \text{— friction-corrected}$$

where n is the number of turns the cord made round the axle during the fall. Where the friction correction is neglected the bracketed factor is dropped, and the result is an upper estimate. (The underlying energy statement: $mgh = \tfrac{1}{2}mv^2 + \tfrac{1}{2}I\omega^2 +$ friction work; with $v = 2h/t$ for uniform acceleration from rest and $\omega = v/r$.)

2. Apparatus and procedure

- **Apparatus:** flywheel on a horizontal axle, cord, mass hanger with slotted masses, stopwatch, metre rule, chalk mark on the wheel's rim.
- Measure the axle radius r (vernier calipers, several orientations). Set the mass so the cord unwraps exactly as the mass reaches the floor/bench stop; measure the fall height h. Count n, the turns of cord on the axle.
- Release; start the watch at release, stop it the instant the cord leaves the axle. Immediately begin counting the wheel's further revolutions N against the chalk mark until it stops.
- Repeat for two or three masses; compute I each time and average.

3. Worked calculation (model data, friction factor shown separately)

RESULT

m = 1.00 kg; r = 0.0200 m; h = 1.00 m; t = 12.0 s; n = 8 turns on the axle; coast-down N = 120 revolutions.

Uncorrected: $I_0 = m r^2 (gt^2/2h - 1) = 1.00 \times (0.0200)^2 \times (9.8 \times 144/2 - 1) = 4.00 \times 10^{-4} \times 704.6 = 0.282$ kg m².

Friction factor: $N/(N + n) = 120/128 = 0.9375$.

$I = 0.282 \times 0.9375 =$ **0.264 kg m²**.

The check on plausibility: the wheel reaches $\omega = (2h/t)/r = 0.1667/0.02 = 8.3$ rad s⁻¹ at release, and its kinetic energy $\frac{1}{2}I\omega^2 = 0.5 \times 0.264 \times 69.4 \approx 9.2$ J, against mgh = 9.8 J — the small remainder is the mass's own kinetic energy and the friction work, as it should be.

4. Precautions

- The mass must detach cleanly (cord loop slips off its peg) — a mass that keeps swinging steals energy and spoils the coast count.
- Count N carefully from the moment of detachment; a second observer counting aloud is standard practice.
- Wind the cord evenly without overlap; overlapping turns change the effective r during the fall.
- The fall should be slow (several seconds): too heavy a mass makes t short, timing error large, and the "uniform acceleration" assumption poor.

5. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. Write the energy balance of the fall and identify each term physically.
2. With m = 0.50 kg, r = 0.0200 m, h = 1.00 m, t = 16.5 s, compute the uncorrected I.
3. What does the coast-down count N measure, and why does the correction factor contain N/(N + n)?
4. Why is the radius of the axle, not of the wheel, the r in the formula?
5. A heavier mass gives a shorter fall time. Why is "as heavy as possible" bad advice here?
6. State two reasons the measured I is an experimental quantity rather than something to compute from the wheel's dimensions in this experiment.

Answer key

1. mgh (potential energy lost) = ½mv² (the falling mass's kinetic energy) + ½Iω² (the wheel's rotational kinetic energy) + work done against friction at the bearings during the fall. The experiment rearranges this balance to expose I.

2. $gt^2/2h = 9.8 \times 272.25/2 = 1334$; $I_0 = 0.50 \times 4.00 \times 10^{-4} \times (1334 - 1) = 2.00 \times 10^{-4} \times 1333 = 0.267$ kg m².
3. N measures how much rotation the bearing friction needs to absorb the wheel's kinetic energy — it calibrates the friction work per revolution. During the fall the wheel makes n turns against the same friction, so the friction's share is scaled by comparing n with N, giving the factor N/(N + n).
4. Because the cord — and therefore the driving torque and the velocity relation $v = \omega r$ — acts at the axle's surface. The wheel's rim radius never enters the energy conversion; only the axle radius converts between the mass's linear motion and the wheel's rotation.
5. Because the timing error is a fixed fraction of a second: shortening t from 12 s to 2 s multiplies the relative timing error sixfold, and t enters the formula squared. A moderate mass keeps t long and the acceleration gentle.
6. The wheel is not a perfect annulus of known density — hub, spokes and rim distribute mass in a way dimensions alone do not fix — and bearing friction is an apparatus property that must be measured in the same run, which is exactly what the method does.
